

BLOCKCHAIN ENGINEERING

ASSIGNMENT - 011

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Batch: 05

Code -

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Assignment-11.py > ...
1  from tkinter import *
2  import time
3  # Simulated blockchain storage
4  blockchain_storage = {
5      "value": None,
6      "history": []
7  }
8
9  # Function to simulate storing a value in the smart contract
10 def store_value():
11     value = entry.get()
12     if value:
13         # Simulate a transaction
14         tx = {
15             "value": value,
16             "time": time.strftime("%Y-%m-%d %H:%M:%S")
17         }
18         blockchain_storage["value"] = value
19         blockchain_storage["history"].append(tx)
20         status_label.config(text="Value stored successfully!")
21         hash_label.config(text=f"Transaction Hash: {hash(str(tx))}")
22         update_history()
23     else:
24         status_label.config(text="Please enter a value to store.")
25
26 # Function to simulate retrieving the stored value
27 def get_value():
28     value = blockchain_storage["value"]
29     if value is not None:
30         result_label.config(text=f"Stored Value: {value}")
31     else:
32         result_label.config(text="No value stored yet.")
33 # Function to update the transaction history display
34 def update_history():
35     history_text = "Transaction History:\n"
36     for tx in blockchain_storage["history"]:
37         history_text += f"{tx['time']}: Stored {tx['value']}\n"
38     history_label.config(text=history_text)
39
40 # GUI setup
41 root = Tk()
42 root.title("Simple Storage Contract Simulation")
43 entry = Entry(root, width=20)
44 entry.pack(pady=10)
45 store_button = Button(root, text="Store Value", command=store_value)
46 store_button.pack(pady=5)
47 get_button = Button(root, text="Get Stored Value", command=get_value)
48 get_button.pack(pady=5)
49 status_label = Label(root, text="")
50 status_label.pack(pady=5)
```

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49 hash_label = Label(root, text="")
50 hash_label.pack(pady=5)
51 result_label = Label(root, text="")
52 result_label.pack(pady=5)
53 history_label = Label(root, text="Transaction History:")
54 history_label.pack(pady=10)
55 root.mainloop()
56 from tkinter import *
57 import time
58

```

Output -



